

Started on	Wednesday, 13 November 2024, 1:02 PM
State	Finished
Completed on	Wednesday, 13 November 2024, 1:19 PM
Time taken	17 mins 23 secs
Marks	2.00/3.00
Grade	2.00 out of 3.00 (66.68%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Incorrect

Mark 0.00 out of 1.00

Let A and X be the sets

$$A = \{a, b, c, d, e\}$$

$$X = \{!, *, +, -\}.$$

In this problem we consider words written in the alphabet which is the union of the two sets: $A \cup X$. This means that a word in this alphabet can contain letters from both A and X .

However, not all words are valid. The following rules apply:

1. A word must start with a letter from the set A .
2. A word must end with a letter from the set A .
3. A word can not contain two adjacent letters from the set X .

For example the word **a+baa!c** is valid, but the words **a+baa+!c**, **!aaa+c** and **acca!** are not.

The length of a word is the number of letters it consists of. For example, the length of **a+baa!c** is 7.

Let a_n equal the number of valid words of length n .

a) Find a recurrence relation satisfied by a_n :

Your last answer was interpreted as follows:

$$a_n = A \cdot n + B \cdot n$$

The variables found in your answer were: $[A, B, a_n, n]$

Syntax help: To express the symbol " a_n " you should enter **a[n]**. For example, if you think the correct answer is $a_{n+1} - 2a_{n-3} = 1$, you should answer **a[n+1] - 2*a[n-3] = 1**.

b) Compute the value

$a_1 =$

Your last answer was interpreted as follows:

9

c) Compute the value

$a_3 =$

Your last answer was interpreted as follows:

27

d) Find the general solution to the recurrence relation satisfied by a_n .

$$a_n = A \cdot n + B \cdot n$$

Your last answer was interpreted as follows:

$$A \cdot n + B \cdot n$$

The variables found in your answer were: $[A, B, n]$

Syntax help: You are free to use constants in your answer. For example, if you think the correct answer is $2 \cdot n - A \cdot 3^n + B$, you should answer **$2 \cdot n - a \cdot 3^n + B$** .

Question 2

Correct

Mark 1.00 out of 1.00

Let $(D_3, \cdot)$ be the dihedral group of symmetries of the regular triangle. We label the vertices of the triangle 1, 2, 3 in a counter-clockwise manner. Then r_1 is the rotational symmetry by 120 degrees which sends vertex 1 to vertex 2, and r_2 is rotation by 240 degrees. The element t_1 is the reflection symmetry which fixes vertex 1, and likewise for t_2 and t_3 .

The multiplication table for this group looks like

id	r_1	r_2	t_1	t_2	t_3
r_1	r_2	id	t_2	t_3	t_1
r_2	.	.	t_3	t_1	.
t_1	.	.	id	r_2	.
t_2	.	.	r_1	id	.
t_3	t_2	t_1	r_2	r_1	id

However, we have left out 9 entries in the places marked by dots.

Find the entire multiplication table for D_3 .

$t_1 \cdot t_3 =$	r1
$r_2 \cdot t_3 =$	t2
$t_2 \cdot t_3 =$	r2
$t_1 \cdot r_2 =$	t2
$r_2 \cdot r_2 =$	r1
$t_2 \cdot r_2 =$	t3
$t_1 \cdot r_1 =$	t3
$r_2 \cdot r_1 =$	id
$t_2 \cdot r_1 =$	t1

Your last answer was interpreted as follows:

r_1

Your last answer was interpreted as follows:

t_2

Your last answer was interpreted as follows:

r_2

Your last answer was interpreted as follows:

t_2

Your last answer was interpreted as follows:

r_1

Your last answer was interpreted as follows:

t_3

Your last answer was interpreted as follows:

t3

Your last answer was interpreted as follows:

id

Your last answer was interpreted as follows:

t1

Question **3**

Correct

Mark 1.00 out of 1.00

a) Let $G = (\mathbb{Z}/2)^{\times 5}$, and let $H \subset G$ be the subgroup

$$H = \{[0, 0, 0, 0, 0], [0, 0, 0, 0, 1], [0, 1, 1, 0, 0], [0, 1, 1, 0, 1]\}.$$

What is the index of $H \subset G$?

$$[G : H] = \boxed{8}$$

Your last answer was interpreted as follows:

8

b) Let D_4 be the dihedral group of order 8 of symmetries of the unit square centered at the origin of $\mathbb{R}^2$. Let $\sigma \in D_4$ be the symmetry that rotates -180 degrees around the origin.

Compute the index of the subgroup generated by σ :

$$[D_4 : \langle \sigma \rangle] = \boxed{4}$$

Your last answer was interpreted as follows:

4

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 200

Signing code:

Your last answer was interpreted as follows:

14458

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

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